
DSC5001: Statistical Machine Learning I

Tutorial 5: Polynomial and Spline Regression

Kangning Cui

City University of Hong Kong (Dongguan)

Semester A, 2026/27

Tutorial Setup

Complete Tasks 1–3.

Task 4 is optional.

- ▶ **Task 1:** Bias–variance and prediction error
- ▶ **Task 2:** Cubic spline and B-spline calculations
- ▶ **Task 3:** Cross-validation and model selection
- ▶ **Task 4:** Python comparison of polynomial and spline models

For Tasks 1–3, show the main calculations and explain the statistical meaning of the result.

Task 1: Bias–Variance and Prediction Error

At a fixed input x_0 , suppose

$$f(x_0) = 5, \quad \sigma^2 = 1.$$

Two learning procedures have the following repeated-sample properties:

	$\mathbb{E}_D[\hat{f}_D(x_0)]$	$\text{Var}_D[\hat{f}_D(x_0)]$
Model A	4.5	0.5
Model B	5.2	1.4

Answer the following:

1. Compute the bias and squared bias of each model.
2. Compute the expected squared prediction error of each model.
3. Which model has the smaller expected prediction error?
4. If Model B has lower training error, does that change your answer to part 3? Explain.

Solution: Task 1

The bias is

$$\text{Bias}[\hat{f}_D(x_0)] = \mathbb{E}_D[\hat{f}_D(x_0)] - f(x_0).$$

Therefore,

$$\text{Bias}_A = 4.5 - 5 = -0.5, \quad \text{Bias}_A^2 = 0.25,$$

and

$$\text{Bias}_B = 5.2 - 5 = 0.2, \quad \text{Bias}_B^2 = 0.04.$$

Using

$$\mathbb{E}[(Y_0 - \hat{f}_D(x_0))^2] = \sigma^2 + \text{Bias}^2 + \text{Var},$$

we obtain

$$R_A = 1 + 0.25 + 0.5 = \boxed{1.75},$$

$$R_B = 1 + 0.04 + 1.4 = \boxed{2.44}.$$

Hence Model A has the smaller expected prediction error.

Lower training error for Model B does not change this conclusion: training fit and expected new-data error are different quantities.

Task 2: Cubic Spline and B-Spline Basis

Consider cubic splines on $[0, 2]$ with one interior knot $\xi = 1$. Here $K = 1$, so $M = K + 4 = 5$. The truncated-power basis is

$$\phi_T(x) = [1, x, x^2, x^3, (x - 1)_+^3]^\top.$$

A B-spline basis for the same spline space is obtained by

$$B(x) = C\phi_T(x), \quad C = \begin{bmatrix} 1 & -3 & 3 & -1 & 1 \\ 0 & 3 & -9/2 & 7/4 & -2 \\ 0 & 0 & 3/2 & -1 & 2 \\ 0 & 0 & 0 & 1/4 & -2 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

- (a) Compute $B(0.5)$ and $B(1.5)$. Which basis functions are zero at each input?
- (b) Explain briefly how the results in (a) illustrate **local support**.
- (c) Let

$$\theta = [1, 2, 4, 6, 3]^\top, \quad f(x) = \theta^\top B(x).$$

Find the equivalent truncated-power coefficient vector β such that

$$f(x) = \beta^\top \phi_T(x).$$

- (d) Verify at $x = 0.5$ that the two representations give the same prediction.

Solution: Task 2 – Basis Functions

From the first row of C ,

$$B_1(x) = 1 - 3x + 3x^2 - x^3 + (x - 1)_+^3.$$

From the fifth row,

$$B_5(x) = (x - 1)_+^3.$$

For $x \leq 1$,

$$B_1(x) = (1 - x)^3, \quad B_5(x) = 0.$$

For $x > 1$,

$$B_1(x) = 0, \quad B_5(x) = (x - 1)^3.$$

Thus the change of basis produces functions whose effects are localized to different parts of the input range.

Solution: Task 2 – Evaluate the Basis

At $x = 0.5$,

$$\phi_T(0.5) = \begin{bmatrix} 1 \\ 1/2 \\ 1/4 \\ 1/8 \\ 0 \end{bmatrix}.$$

Multiplying by C gives

$$B(0.5) = \begin{bmatrix} 1/8 \\ 19/32 \\ 1/4 \\ 1/32 \\ 0 \end{bmatrix}.$$

Solution: Task 2 – Evaluate the Basis

At $x = 1.5$,

$$\phi_T(1.5) = \begin{bmatrix} 1 \\ 3/2 \\ 9/4 \\ 27/8 \\ 1/8 \end{bmatrix},$$

so

$$\boxed{\mathbf{B}(1.5) = \begin{bmatrix} 0 \\ 1/32 \\ 1/4 \\ 19/32 \\ 1/8 \end{bmatrix}}.$$

Notice that

$$B_5(0.5) = 0, \quad B_1(1.5) = 0.$$

This illustrates the local-support behavior of the B-spline basis.

Solution: Task 2 – B-Spline Prediction

Using

$$f(x) = \boldsymbol{\theta}^\top \mathbf{B}(x), \quad \boldsymbol{\theta} = [1, 2, 4, 6, 3]^\top,$$

we obtain

$$\begin{aligned} f(0.5) &= 1 \left(\frac{1}{8} \right) + 2 \left(\frac{19}{32} \right) + 4 \left(\frac{1}{4} \right) + 6 \left(\frac{1}{32} \right) \\ &= \boxed{\frac{5}{2}}. \end{aligned}$$

Similarly,

$$\begin{aligned} f(1.5) &= 2 \left(\frac{1}{32} \right) + 4 \left(\frac{1}{4} \right) + 6 \left(\frac{19}{32} \right) + 3 \left(\frac{1}{8} \right) \\ &= \boxed{5}. \end{aligned}$$

Solution: Task 2 – Same Spline, Different Basis

Since

$$B(x) = C\phi_T(x),$$

we have

$$\begin{aligned} f(x) &= \theta^\top B(x) \\ &= \theta^\top C\phi_T(x) \\ &= (C^\top \theta)^\top \phi_T(x). \end{aligned}$$

Therefore,

$$\beta = C^\top \theta = \begin{bmatrix} 1 \\ 3 \\ 0 \\ 0 \\ -4 \end{bmatrix}.$$

So the same spline is

$$f(x) = 1 + 3x - 4(x - 1)_+^3.$$

Task 3: Cross-Validation and Model Selection

4-fold cross-validation (CV): fit on three folds and evaluate on the remaining fold, rotating the held-out fold. Each validation fold contains 10 observations. The table gives training MSE and validation SSE.

Model	Train MSE	$F1$	$F2$	$F3$	$F4$
Degree 2	2.10	24	20	22	26
Degree 5	1.30	18	25	21	28
Degree 9	0.45	11	30	27	32
Spline, 3 knots	1.15	17	19	18	20

For each fold,

$$\text{MSE}_{\text{fold}} = \frac{\text{SSE}_{\text{fold}}}{10}.$$

- (a) Convert the four SSE values into fold MSEs and compute the mean CV MSE for each model.
- (b) Compute

$$\text{CV gap} = \text{CV MSE} - \text{training MSE}.$$

Which model shows the strongest sign of overfitting?

- (c) Which model should be selected by CV?
- (d) Degree 2 and degree 5 have the same CV MSE. If these were the only two candidates, which would you choose, and why?
- (e) After model selection, how should the final test set be used?

Solution: Task 3 – Fold and CV Errors

Since each validation fold has 10 observations,

Model	$F1$	$F2$	$F3$	$F4$	CV MSE
Degree 2	2.4	2.0	2.2	2.6	2.30
Degree 5	1.8	2.5	2.1	2.8	2.30
Degree 9	1.1	3.0	2.7	3.2	2.50
Spline	1.7	1.9	1.8	2.0	1.85

For example,

$$CV_{\text{Degree 2}} = \frac{2.4 + 2.0 + 2.2 + 2.6}{4} = \boxed{2.30}.$$

The smallest CV MSE is

$\boxed{1.85}$

for the cubic spline with 3 knots.

Solution: Task 3 – Training versus CV Error

Compute

$$\text{CV gap} = \text{CV MSE} - \text{training MSE}.$$

Model	Train MSE	CV MSE	CV gap
Degree 2	2.10	2.30	0.20
Degree 5	1.30	2.30	1.00
Degree 9	0.45	2.50	2.05
Spline	1.15	1.85	0.70

The degree-9 polynomial has

the largest CV gap

despite having the lowest training MSE.

This is the strongest sign of overfitting among the four candidates.

Solution: Task 3 – Model Selection

Using CV error, select

the cubic spline with 3 knots.

It has the smallest mean CV MSE: 1.85.

If only degree 2 and degree 5 were available, both have

$$\text{CV MSE} = 2.30.$$

With essentially equal validation performance, choose the simpler model:

degree 2.

After choosing the model:

select by CV $\rightarrow$ refit on non-test data $\rightarrow$ evaluate once on the test set.

The test set should not be used repeatedly for model selection.

Optional Task 4: Polynomial versus B-Spline in Python

Generate a nonlinear regression dataset using

$$Y = 2 + \sin(3X) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 0.3^2).$$

Compare:

- ▶ degree-3 polynomial regression;
- ▶ degree-7 polynomial regression;
- ▶ cubic B-spline regression with 6 knots.

Use the same 5-fold cross-validation split for all three models.

1. Report the mean cross-validation MSE.
2. Select the model with the smallest CV MSE.
3. Fit all three models to the full dataset.
4. Plot the fitted curves together with the observations.

Use a fixed random seed so the result is reproducible.

Solution: Optional Task 4 – Setup

```
import numpy as np
import matplotlib.pyplot as plt

from sklearn.pipeline import Pipeline
from sklearn.preprocessing import PolynomialFeatures
from sklearn.preprocessing import SplineTransformer
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import KFold, cross_val_score

rng = np.random.default_rng(7)

n = 120
X = rng.uniform(-3, 3, size=(n, 1))
x = X[:, 0]

y = 2 + np.sin(3 * x)
y += rng.normal(0, 0.3, size=n)

cv = KFold(
    n_splits=5,
    shuffle=True,
    random_state=0
)
```

Solution: Optional Task 4 – Models and CV

```
poly3 = Pipeline([
    ("poly", PolynomialFeatures(
        degree=3, include_bias=False)),
    ("lr", LinearRegression())
])
poly7 = Pipeline([
    ("poly", PolynomialFeatures(
        degree=7, include_bias=False)),
    ("lr", LinearRegression())
])
spline = Pipeline([
    ("spline", SplineTransformer(
        n_knots=6,
        degree=3,
        include_bias=False)),
    ("lr", LinearRegression())
])
models = {
    "Polynomial 3": poly3,
    "Polynomial 7": poly7,
    "B-spline": spline
}
```

Solution: Optional Task 4 – Plot

```
for name, model in models.items():
    scores = cross_val_score(
        model, X, y, cv=cv,
        scoring="neg_mean_squared_error"
    )
    print(name, -scores.mean())

grid = np.linspace(-3, 3, 400).reshape(-1, 1)

plt.scatter(X[:, 0], y, s=18)

for name, model in models.items():
    model.fit(X, y)
    pred = model.predict(grid)
    plt.plot(grid[:, 0], pred, label=name)

plt.xlabel("x")
plt.ylabel("y")
plt.legend()
plt.show()
```

Solution: Optional Task 4 – Plot

With the fixed seed, the mean CV MSEs are approximately

Model	Mean CV MSE
Degree-3 polynomial	0.532
Degree-7 polynomial	0.208
Cubic B-spline	0.101

Thus the B-spline has the smallest cross-validation error for this generated dataset.

Tutorial Summary

After this tutorial, you should be able to:

- ▶ use the bias–variance decomposition to compare prediction error;
- ▶ evaluate truncated-power spline features;
- ▶ compute predictions from supplied B-spline basis values;
- ▶ use cross-validation to choose model complexity;
- ▶ compare polynomial and B-spline regression computationally.

Tutorial 5: nonlinear regression, splines, and model selection

Questions and discussion